JUNWOO PARK

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EDUCATION

University of California, Irvine — **Irvine, CA**
M.S. in Computer Science — Sep. 2026 – Jun. 2028 (Expected)

Sungkyunkwan University (SKKU) — **Korea**
B.S. in Computer Science and Engineering, GPA: 4.31/4.50 — Feb. 2026

RESEARCH EXPERIENCE

SKKU AI-Healthcare Lab
Research Assistant (Advisor: Prof. Sujee Lee) — Feb. 2025 – Jul. 2026

- Built a molecular LLM based on Llama 3.1 8B through graph-language alignment and performed instruction tuning using LoRA on GNN-derived structural rationales to improve property prediction [1].
- Developed a multimodal deep learning model that integrates molecular graphs, sequences, images, and spectra using attention mechanisms for toxicity prediction [2].
- Collaborated with Samsung C-Lab on the "Muscle Tracker" project.

PUBLICATIONS

[1] **J. Park**, M. Shin, C. S. Lee, and S. Lee, Multi-Granular Rationale-Guided Molecular LLM for Property Prediction, arXiv:2608.10480, 2026.

[2] **J. Park** and S. Lee, MoltiTox: a multimodal fusion model for molecular toxicity prediction, Frontiers in Toxicology, 2025.

PROJECTS

Reasoning for Embodied Agents, UC Irvine
Student Researcher (Advisor: Dr. Liwei Jiang) — Aug. 2026 – Present

- Develop an egocentric video benchmark to evaluate whether frontier VLMs acting as embodied agents can behave appropriately while carrying out everyday tasks.
- Build a pipeline that extracts behavioral norms from an existing egocentric video dataset to guide embodied agents toward socially appropriate behavior.

Project SKKU Character: AI Platform for University Mascot, SKKU
AI Engineer — Oct. 2025 – Feb. 2026

- Secured 5M KRW (∼3,500 USD) in funding through the university entrepreneurship program.
- Implemented a FastAPI service to process image generation requests via the Replicate API and added a filtering layer to block unsafe prompts before generation.

Muscle Tracker: AI-driven Analysis for Strength Training, SKKU
Research Assistant — Aug. 2025 – Nov. 2025

- Collected and preprocessed EMG and IMU sensor data from 14 participants with Butterworth filtering, interpolation, and root-mean-square (RMS) computation using Python.
- Designed an algorithm to predict maximum muscle activation via sigmoid curve fitting and co-developed a temporal convolutional network for exercise classification, achieving 96% accuracy.

NotiSKKU: Student Announcement App, SKKU
Team Leader Aug. 2024 – Dec. 2025

- Led a 4-member team to create a mobile app aggregating campus announcements, informed by a survey of 60+ students.
- Built the frontend in Flutter and launched the app on Google Play, reaching 200+ downloads.

VisualVroom: Alert System for Hearing-Impaired Drivers, SKKU
Team Leader Apr. 2025 – Aug. 2025

- Directed a team to create a real-time alert system for hearing-impaired drivers.
- Developed a Vision Transformer-based model to classify sound sources and their directions using smartphone audio input, achieving 83.9% accuracy.

TEACHING & MENTORING EXPERIENCE

Course: Basis and Practice in Programming, SKKU
Teaching Assistant Fall 2025

- Guided 60 students in C programming coursework, evaluating assignments and providing feedback.

Software Mentoring Program, SKKU
Mentor May – Sep. 2024, May – Sep. 2025

- Taught Python fundamentals and conducted lab sessions for high school students.

EXTRACURRICULAR ACTIVITIES

Google Developer Groups on Campus, SKKU
President Feb. 2025 – Feb. 2026

- Led a student organization with 20 members and hosted monthly tech talks featuring global tech professionals and researchers.
- Established a partnership with Mission San Jose High School in California; co-hosted a joint programming contest in Feb. 2026.

AWARDS & HONORS

First Place, 2025 Uni-DTHON (Hackathon) Nov. 2025

2025 President's List, Sungkyunkwan University Sep. 2025

Grand Prize, ICT Award Korea 2025 Aug. 2025

SKILLS

Languages	Python, C, C++, Java, SQL, HTML/CSS
Frameworks & Tools	PyTorch, FastAPI, Flutter, Git, Docker, Linux, MATLAB